

Evaluating and Improving Cross-Dataset Generalization of Machine-Learning-Based Intrusion Detection Systems



Figure 1: NSUT logo

B.Tech. Project – I (Mid-Semester Report)

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1. Abstract

Machine-learning (ML) based Network Intrusion Detection Systems (NIDS) routinely report near-perfect accuracy, but these numbers are almost always obtained by training and testing on the *same* dataset. When such a model is deployed on a different network, its performance frequently collapses — in some studies to below 40%, or even to random guessing — because it has learned dataset-specific artefacts (IP addresses, port numbers, collection-window statistics) rather than genuine attack behaviour. This project studies and attempts to reduce this **cross-dataset generalisation gap** across three structurally different benchmark datasets: **CIC-IDS-2017**, **UNSW-NB15**, and **TON_IoT**. Our key idea is to use **metaheuristic / bio-inspired optimisation** (Genetic Algorithm, Particle Swarm Optimization, and other evolutionary/swarm methods) to select the small set of **shared features that remain robust across all three network environments**, rather than optimising for accuracy on a single dataset as all prior metaheuristic NIDS work does. We first build a *common semantic feature subspace* by harmonising the three datasets, then evaluate multiple ML models (Random Forest, XGBoost/LightGBM, SVM, Logistic Regression, and a small MLP) under a strict train-on-source / test-on-target protocol, and finally optimise the feature subset using a **cross-dataset fitness function**. This mid-semester report presents the completed dataset analysis, the literature foundation, the problem formulation, and the detailed methodology and plan; model benchmarking and optimisation form the remaining work.

Keywords: intrusion detection, cross-dataset generalisation, metaheuristic optimisation, feature selection, CIC-IDS-2017, UNSW-NB15, TON_IoT.

2. Introduction

2.1 Background

A Network Intrusion Detection System (NIDS) monitors network traffic to detect malicious activity. Traditional signature-based systems can only detect known attacks, so research has increasingly turned to **machine learning**, which can learn patterns of malicious behaviour from labelled traffic and detect both known and some unknown attacks. Public benchmark datasets — CIC-IDS-2017 [5], UNSW-NB15 [6], and TON_IoT [7] — have enabled rapid progress in this area.

2.2 The problem with standard evaluation

The dominant evaluation practice in the literature is to split **one** dataset into training and testing portions. Models then report accuracy above 99%. However, this setting is optimistic: the training and test data come from the *same* network, the *same* sensors, and the *same* collection period. In the real world, an IDS is usually trained on one network (or on a vendor’s dataset) and then deployed on a *different* network with different topology, different services, and different traffic statistics. Under this realistic setting, the literature shows that performance collapses [1], [2].

2.3 Cross-dataset generalisation

Cross-dataset generalisation asks: *how well does a model trained on dataset A perform on dataset B?* The answer in the literature is consistently poor. Cantone et al. [1] found that cross-dataset accuracy is close to random chance for most combinations of fourth-generation datasets. Hossain et al. [2] found that a Random Forest reaching 95–99% within UNSW-NB15 or TON_IoT drops to below 40% when transferred between them. The root cause is that different datasets use different **feature extraction tools** (CICFlowMeter vs Argus/Bro vs Zeek) and contain features that do not mean the same thing across networks.

2.4 This project

This project (i) **measures** the cross-dataset generalisation of several ML models across CIC-IDS-2017, UNSW-NB15 and TON_IoT; (ii) **characterises** which features and attack classes transfer and which are dataset-specific artefacts; and (iii) **improves** generalisation using **meta-heuristic feature selection** whose objective is cross-dataset robustness. Our novelty is the combination of *cross-dataset evaluation* with *metaheuristic feature selection optimised for transfer*, which — to our knowledge — has not been attempted in the NIDS literature.

3. Motivation

1. **Deployment reality.** Enterprises rarely have large labelled datasets for their own network. They must reuse models trained elsewhere. A model that fails to transfer is useless in practice, regardless of its in-dataset accuracy.
2. **The gap is large and well-documented.** Independent studies [1]–[3] show the gap is not a minor degradation but a collapse. Fixing even part of it has real security value.

3. **Feature selection is the natural lever.** Cross-dataset failure is largely a *feature* problem (irrelevant or dataset-specific features mislead the model). The metaheuristic feature-selection literature is mature [8]–[12] — but universally optimises **same-dataset** accuracy. Re-aiming the objective at **transfer** is a simple, powerful and unexplored idea.
 4. **Bio-inspired optimisation is well-suited.** The search space of feature subsets is combinatorial and the fitness landscape is non-convex; genetic and swarm algorithms (GA, PSO, DE) are designed for exactly this kind of problem.
 5. **Practical output.** The project produces not just a model but *guidelines* about which feature families are safe to rely on across networks — directly useful to practitioners.
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4. Literature Survey

We surveyed 15 core papers, grouped into five themes. Full details (problem, method, datasets, results, limitation) are given in the companion document `02_literature_review.md`; the essential findings are summarised here.

4.1 Cross-dataset generalisation (the core problem)

- **Cantone et al. (2024) [1]** built a cross-dataset benchmark over four datasets using a restricted common feature set, and showed that cross-dataset accuracy is near random chance for most pairs, with failures caused by dataset-specific artefacts. This is our closest prior work, but it is **diagnostic only** — it proposes no mitigation.
- **Hossain et al. (2026) [2]** evaluated RF, Logistic Regression and Naïve Bayes on UNSW-NB15 $\leftrightarrow$ TON_IoT. In-dataset accuracy was 95.08% and 99.79%, but cross-dataset accuracy fell **below 40%** in both directions.
- **Hakim et al. (2026) [3]** showed lightweight IIoT models rely on a **port-category shortcut** (present 96–435× more often in source attacks than in targets), and that the evaluation protocol can *reverse* conclusions.
- **Butt et al. (2026) [4]** found that tabular representation learning can transfer *sometimes*, but performance depends strongly on the source–target pair.

Synthesis: cross-dataset evaluation is well studied, but the field stops at diagnosis.

4.2 Benchmark datasets

- **CIC-IDS-2017 [5], UNSW-NB15 [6] and TON_IoT [7]** are the three most-used modern NIDS benchmarks; they come from different networks and different feature extractors. Ring et al.’s survey [14] explains why such heterogeneity is unavoidable. Vitorino et al. [15] showed that the well-known flaws in CICIDS2017 can materially change model rankings, motivating careful cleaning.

4.3 Metaheuristic / bio-inspired feature selection

- **Fatima & Ali (2022) [8]** used a wrapper **Genetic Algorithm** with stacking/bagging ensembles on UNSW-NB15 and CICDDoS2019 — strong results, but fitness is **same-dataset**.

- **Suman et al. (2019) [9]** used **NSGA-II** multi-objective optimisation of information-theoretic measures for unsupervised feature selection on KDD-99/NLS-KDD/Kyoto.
- **Li (2026) [10]** proposed **MPDGGA**, a multi-population diversity-guided GA, achieving the best accuracy on 10/11 datasets with very few features.
- **Mojtahedi et al. (2022) [11]** combined **Whale Optimization + GA** with KNN.
- **Yang et al. (2026) [12]** used **NSGA-III** to jointly optimise features and hyper-parameters under efficiency objectives.

Synthesis: metaheuristic feature selection is effective and mature, but **every** work we found optimises for *same-dataset* accuracy.

4.4 Domain adaptation / transfer

- **Layeghy et al. (2022/23) [13]** (DI-NIDS) learn **domain-invariant features** via adversarial adaptation and detect anomalies with a One-Class SVM, improving cross-domain performance — but this needs aligned feature spaces and target-domain data.

4.5 Literature summary table

#	Title	Author(s)	What it does	Conclusion
1	Cross-Dataset Generalization of ML for NIDS	Cantone et al., 2024	Cross-dataset benchmark, common features	Near-chance cross-dataset; diagnosis only
2	Assessing Generalisation Capability	Hossain et al., 2026	RF/LR/NB on UNSW $\leftrightarrow$ TON	<40% cross-dataset vs 95–99% in-dataset
3	Cross-Domain Failure in Lightweight IDS	Hakim et al., 2026	Lightweight models on 3 IIoT sets	Port shortcut; protocol reverses results
4	Tabular Representation Learning for NIDS	Butt et al., 2026	Representation learning + transfer	Transfer depends on pair
5	CIC-IDS-2017	Sharafaldin et al., 2018	Dataset	Primary source
6	UNSW-NB15	Moustafa & Slay, 2015	Dataset	Second dataset
7	TON_IoT	Moustafa, 2021	Dataset	Third dataset (Zeek)
8	Metaheuristic classifiers for ID	Fatima & Ali, 2022	GA feature selection + ensembles	Strong single-dataset; no transfer
9	Unsupervised FS via NSGA-II	Suman et al., 2019	Multi-objective FS	Pareto subsets; single-dataset
10	MPDGGA	Li, 2026	Multi-population GA FS	Best 10/11 datasets; single-dataset
11	GA + Whale Optimization FS	Mojtahedi et al., 2022	Hybrid bio-inspired FS	Swarm hybridisation viable
12	NSGA-III AutoML IDS	Yang et al., 2026	Multi-objective AutoML	Efficiency, not transfer
13	DI-NIDS	Layeghy et al., 2022	Domain-invariant features	Better cross-domain; needs target data
14	Survey of NIDS Data Sets	Ring et al., 2019	Dataset taxonomy	Explains heterogeneity
15	Adversarial Robustness (New-CICIDS)	Vitorino et al., 2024 7	Robustness on corrected CICIDS2017	Corrected data changes conclusions

4.6 Identified gap

Group	Cross-dataset eval?	Metaheuristic FS?	Optimised for transfer?
Diagnostic studies [1]–[4]	Yes	No	No
Metaheuristic FS [8]–[12]	No	Yes	No (same-dataset fitness)
Domain adaptation [13]	Yes	No	No (needs target data)
This project	Yes (3 datasets)	Yes (GA, PSO, ...)	Yes

No published work performs metaheuristic feature selection using a **cross-dataset generalisation objective** for NIDS. This is the central gap our project addresses.

5. Problem Statement

Existing ML-based intrusion detection systems are evaluated almost exclusively with same-dataset train/test splits, which overstate their real-world usefulness. When such models are deployed on a different network — a *structurally different* dataset with a different feature extractor, topology and attack distribution — their detection performance degrades sharply, often to near-random levels [1], [2]. This cross-dataset generalisation gap is caused largely by reliance on dataset-specific features (identity fields such as IP addresses and ports, and collection-dependent statistics) rather than genuine, transferable attack behaviour. Feature-selection methods that could address this are optimised only for same-dataset accuracy [8]–[12]. **There is therefore a need to (a) quantitatively characterise the cross-dataset generalisation gap across structurally different NIDS benchmarks, and (b) develop a feature-selection approach, driven by bio-inspired optimisation, whose objective is cross-dataset robustness rather than single-dataset accuracy.**

6. Objectives

1. **Evaluate** the cross-dataset generalisation of multiple ML/DL models trained on one dataset and tested on structurally different ones (CIC-IDS-2017, UNSW-NB15, TON_IoT), in addition to same-dataset splits.
2. **Characterise** the sources of the generalisation gap — which attack categories, features and traffic patterns hold up across environments versus which are dataset-specific artefacts.
3. **Explore and compare** metaheuristic/bio-inspired optimisation techniques (Genetic Algorithm, Particle Swarm Optimization, and additional swarm/evolutionary methods) for selecting features and tuning models **specifically for cross-dataset robustness**.

4. **Benchmark** the optimised approach against standard feature-selection and tuning baselines (mutual information, random-forest importance, PCA, RFE, LASSO) to determine which techniques most effectively narrow the gap.
5. **Consolidate** the findings into practical guidelines for building IDS that generalise reliably to previously unseen network environments.

7. Proposed Methodology

The project proceeds in seven phases (Figure 1).

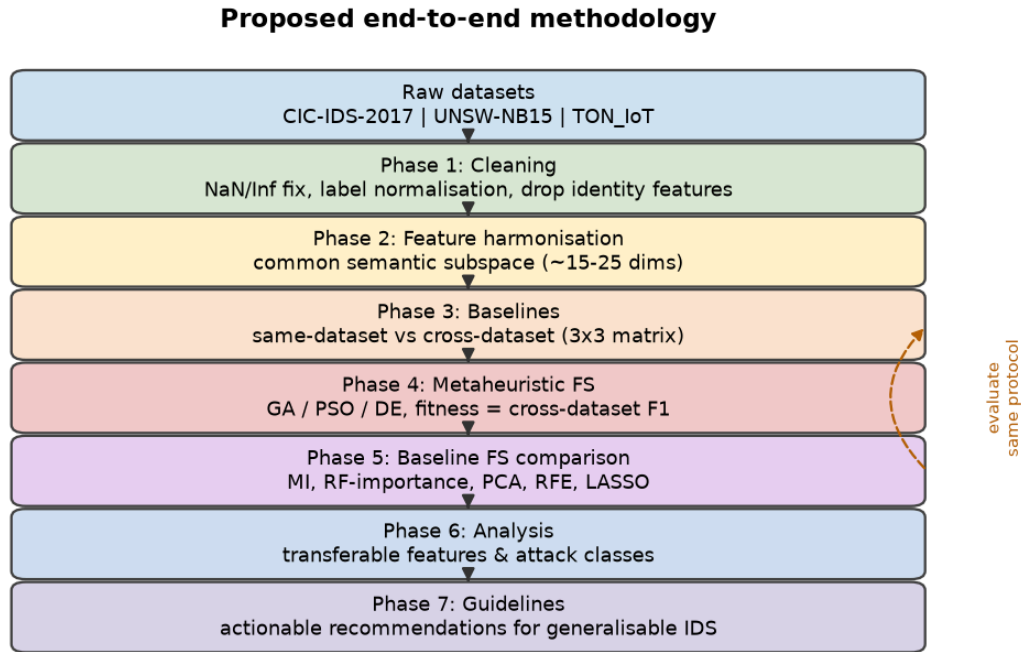


Figure 2: Proposed end-to-end methodology

Figure 1: Proposed methodology pipeline.

7.1 Phase 1 — Data cleaning and preprocessing

- Strip whitespace from headers and remove the duplicated Fwd Header Length column in CIC-IDS-2017.
- Replace Infinity values (Flow Bytes/s, Flow Packets/s) and impute/drop the small number of NaN cells.
- Normalise labels (including the non-ASCII en-dash in Web Attack – *) and map each dataset’s labels to (i) a binary benign/attack target and (ii) a coarse multi-class taxonomy.
- **Drop identity/leakage features** (CIC-IDS-2017 Destination Port; TON_IoT src_ip, dst_ip, src_port, dst_port; UNSW-NB15 id).
- Encode categoricals and standardise numeric features using per-dataset scalers.

7.2 Phase 2 – Feature harmonisation (the shared subspace)

Because the three datasets use different extractors, we construct a **common semantic feature subspace** by mapping semantically equivalent fields (Table below). Figure 2 shows how few features are actually shared.

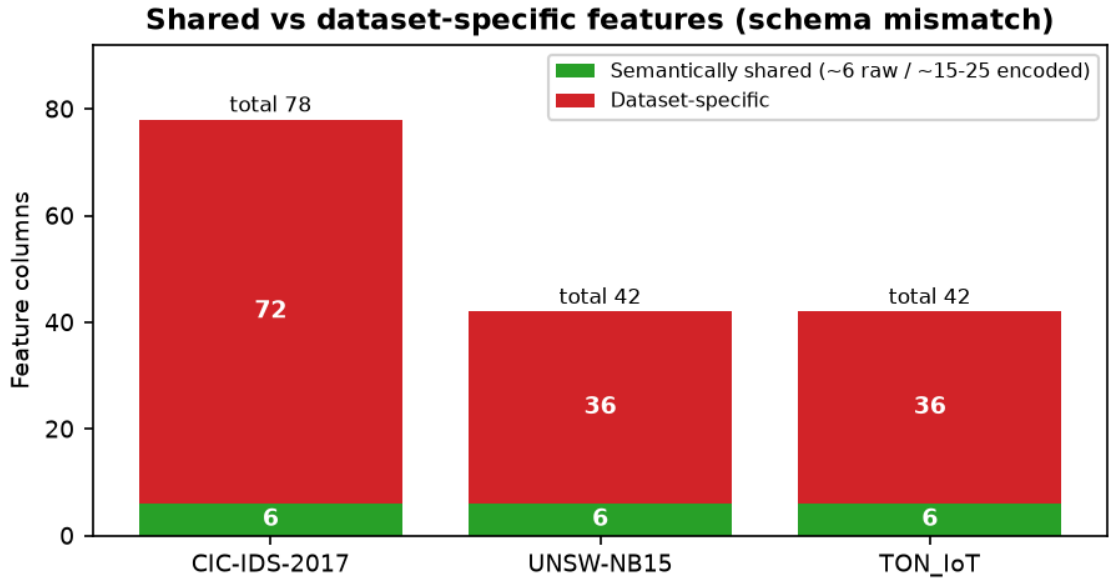


Figure 3: Shared subspace

Figure 2: Few features are semantically shared across the three datasets.

Semantic concept	CIC-IDS-2017	UNSW-NB15	TON_IoT
Duration	Flow Duration	dur	duration
Source bytes	Total Length of Fwd Packets	sbytes	src_bytes
Destination bytes	Total Length of Bwd Packets	dbytes	dst_bytes
Source packets	Total Fwd Packets	spkts	src_pkts
Destination packets	Total Backward Packets	dpkts	dst_pkts
Protocol (encoded)	(implied)	proto	proto
Service (encoded)	Destination Port	service	service
State (encoded)	(TCP flags)	state	conn_state
1. derived ratios (bytes/packet, packets/s, fwd/ bwd ratio)			

After encoding the categorical fields and adding derived rates/ratios, the shared subspace is roughly **15–25 dimensions**. This is the space our optimiser searches.

7.3 Phase 3 — Baseline benchmark (measure the gap)

- **Models:** Random Forest, XGBoost/LightGBM, SVM (RBF), Logistic Regression, and a small MLP.
- **Same-dataset protocol:** stratified 5-fold cross-validation.
- **Cross-dataset protocol:** train on one dataset, test on another, for all ordered pairs (a 3×3 source→target matrix; the diagonal is same-dataset).
- **Metrics:** balanced accuracy, macro-F1, per-class F1, ROC-AUC, reported as mean ± std over multiple seeds. *Accuracy alone is not used* because the datasets have very different class imbalances.

7.4 Phase 4 — Metaheuristic feature selection for transfer (core contribution)

- A candidate solution is a **binary mask** over the shared subspace.
- **Fitness = cross-dataset generalisation:** a fast model (RF/XGB) is trained on the selected features of the source and evaluated on the target(s); fitness is the average balanced accuracy / macro-F1 across source→target pairs. Crucially, the optimiser is rewarded for *transfer*, not for fitting the source.
- **Optimisers:** Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Differential Evolution or Whale Optimization, each run multiple times with different seeds.

- Optional **multi-objective** variant (NSGA-II) trading cross-dataset F1 against the number of features.

7.5 Phase 5 — Baselines for comparison

The same cross-dataset protocol is applied to standard feature-selection baselines — mutual information, random-forest importance, PCA, recursive feature elimination, and LASSO — each matched to the same subset size as the metaheuristic. The key question: **does the transfer-optimised subset beat same-dataset-optimised subsets when both are deployed cross-dataset?**

7.6 Phase 6 — Analysis and characterisation

Feature-importance/SHAP analysis, per-class confusion between source and target, t-SNE/UMAP visualisation of domain shift, and ablations (with/without leakage features; balanced vs natural class distributions; subset size vs performance).

7.7 Phase 7 — Guidelines

Distil results into practical recommendations about which feature families are safe to rely on across networks.

7.8 Scope and evaluation protocol decisions

Decision	Choice
Classification	Binary (benign/attack) primary; coarse multi-class secondary
Models	Classic ML (RF, XGB/LGBM, SVM, LR) + small MLP
Optimisation target	Feature subset (+ light hyper-parameter tuning)
Feature space	Shared semantic subspace (~15–25 dims) for cross-dataset runs
Metrics	Balanced accuracy, macro-F1, per-class F1
Leakage handling	Remove IP/port/ID/timestamp features
Reproducibility	Fixed seeds, repeated runs, cached harmonised matrices

8. Implementation

The following have been implemented and verified during this phase.

1. **Environment and repository.** A reproducible Python environment and repository were set up (src/, reports/, with a requirements.txt), and the code is version-controlled.

2. **Dataset acquisition and inspection.** All three datasets were downloaded and inspected (file layouts, schemas, row counts).
3. **Automated dataset-analysis pipeline** (`src/analyze_datasets.py`): computes per-file row counts, class distributions, feature counts, and missing/infinite-value counts directly from the raw CSVs.
4. **Figure-generation pipeline** (`src/generate_report_assets.py`, `src/generate_extra_figures.py`): automatically produces all report figures.
5. **Feature-harmonisation design.** The semantic mapping of Section 7.2 and the binary + coarse label taxonomy have been specified and are ready to be coded.
6. **Cleaning rules.** The exact handling for Inf/NaN values, the duplicated column, the label-encoding artefact, and identity features has been determined.

Remaining implementation: cleaning/harmonisation code, the model benchmark, the meta-heuristic optimisers, the baseline feature selectors, and the analysis suite (Phases 1–6 of the plan).

9. Results and Discussion

9.1 Dataset composition (measured)

Table 1 and Figures 3–5 summarise the measured composition of the three datasets.

Table 1: Measured dataset characteristics.

Property	CIC-IDS-2017	UNSW-NB15	TON_IoT
Total records	2,830,743	257,673	211,043
Feature columns	78 (1 duplicate)	42	42
Label scheme	15 classes	9 attacks + Normal	10 classes
Missing cells	1,358	0	0
Infinite cells	4,376	0	0
Benign fraction	≈80.3%	36.1%	23.7%

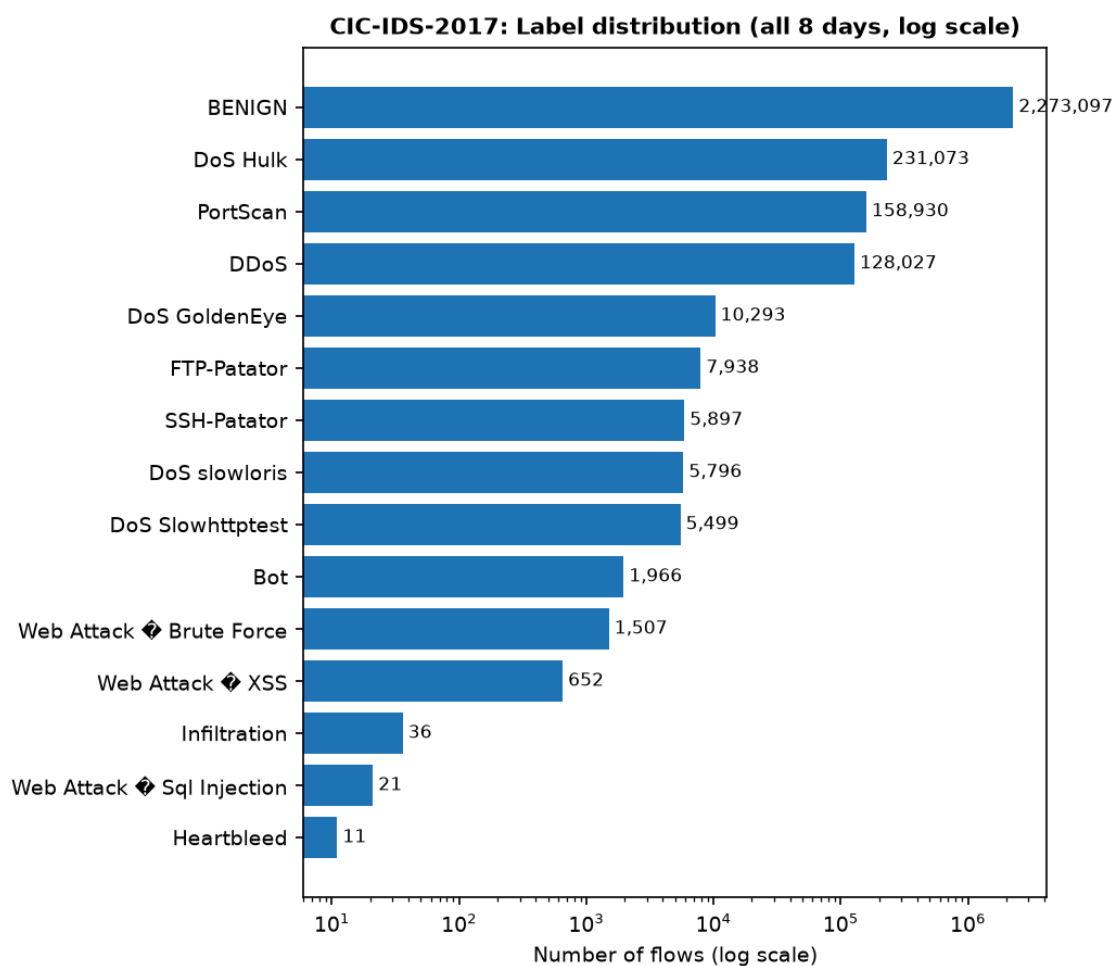


Figure 4: CIC-IDS-2017 class distribution

Figure 3: CIC-IDS-2017 class distribution (log scale) — extreme imbalance.

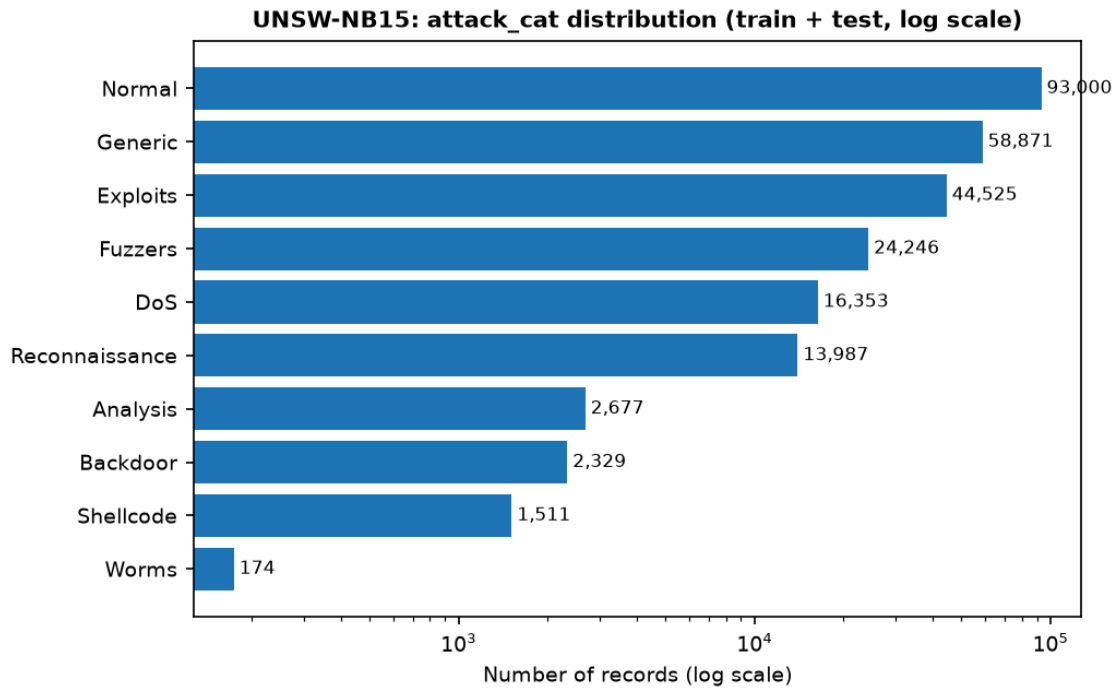


Figure 5: UNSW-NB15 class distribution

Figure 4: UNSW-NB15 class distribution (log scale).

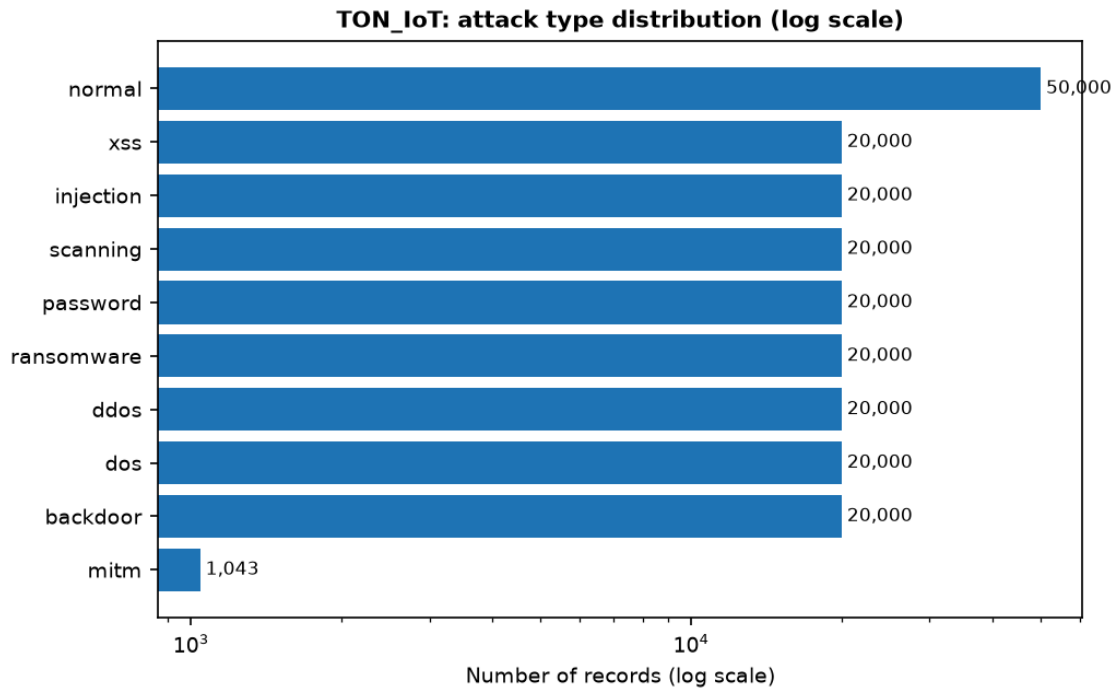


Figure 6: TON_IoT class distribution

Figure 5: TON_IoT class distribution (log scale) — near-balanced attacks.

9.2 Discussion of the dataset analysis

1. **Severe, dataset-specific class imbalance.** CIC-IDS-2017 is about 80% benign and contains classes with only 11 (Heartbleed) and 36 (Infiltration) samples, whereas TON_IoT is almost perfectly balanced (20,000 per attack). This makes **overall accuracy a misleading metric** and confirms our decision to use balanced accuracy and macro-F1.
2. **Data-quality issues.** CIC-IDS-2017 contains 4,376 infinite and 1,358 missing cells (division-by-zero artefacts in Flow Bytes/s and Flow Packets/s) and a duplicated Fwd Header Length column — these must be cleaned before training.
3. **The schema mismatch is the central finding.** The three datasets share **no common column name**, and only about six raw features are semantically shared (Figure 2). This directly explains why models trained on one dataset fail on another: most of their input features simply do not exist, or mean something different, in the target network.
4. **Identity features are present and dangerous.** TON_IoT exposes raw IPs and ports; CIC-IDS-2017 exposes the destination port. If left in, a model can memorise the *attack scenario* instead of learning behaviour, which inflates same-dataset scores and destroys transfer — exactly the “shortcut” effect reported in [3] and the artefact effect in [1].

9.3 Expected model results (baseline, to be completed)

Based on the literature [1], [2], we expect the same-dataset baselines to reach very high scores (≈ 0.95 – 0.99 F1) while cross-dataset performance will drop sharply, sometimes to near chance. These results will form the baseline against which our transfer-optimised feature selection is measured. The **relative improvement over this baseline** — not the absolute cross-dataset accuracy — is the project’s primary success metric.

10. Conclusion and Future Work

This mid-semester report established the foundation of the project. We acquired and analysed three structurally different NIDS benchmarks and showed that they suffer from a fundamental **feature-schema mismatch**, severe dataset-specific class imbalance, and the presence of identity/leakage features — all of which explain the cross-dataset generalisation gap reported in the literature. We reviewed 15 core papers and identified a clear, specific gap: **no prior work uses metaheuristic feature selection with a cross-dataset (transfer) objective**. We formulated five objectives and a seven-phase methodology to fill this gap, and we set up a reproducible codebase and automated analysis/plotting pipeline.

Future work (remaining semester):

1. Implement cleaning and feature harmonisation; cache the harmonised matrices.
2. Run the same-dataset and cross-dataset baseline benchmark (3×3 matrix) for all models.
3. Implement GA, PSO and a third metaheuristic with the cross-dataset fitness function.
4. Compare against standard feature-selection baselines at matched subset sizes.
5. Analyse transferable vs dataset-specific features and attack classes; perform ablations.

6. Consolidate practical guidelines and prepare the final report.

Gantt chart:

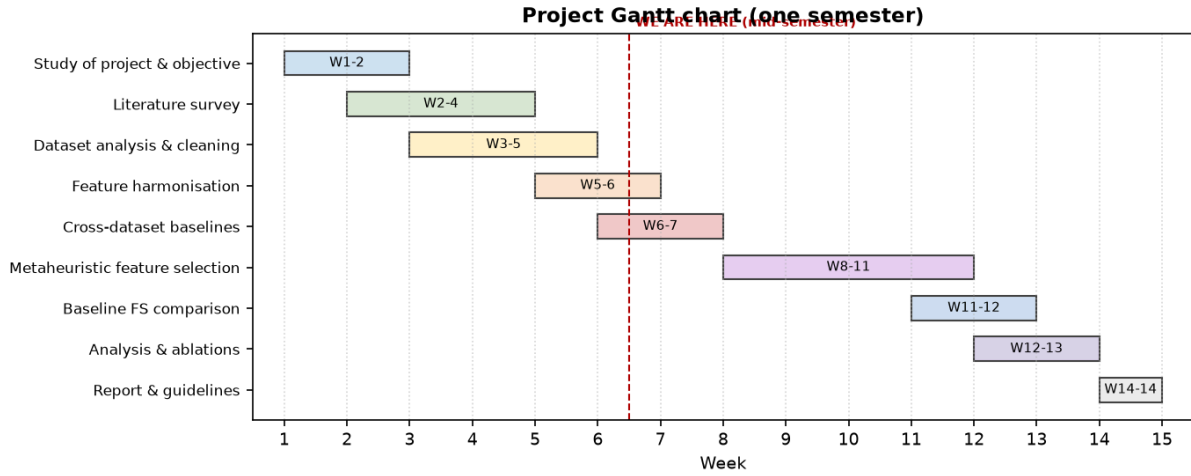


Figure 7: Gantt chart

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